Register spill study: status and percentage tables

L1D 4 KiB, L1I 32 KiB, L2 32 KiB, 64 B lines, LRU · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

Completion

set	config	completed	running	pending	other	remaining
test input, full runs	small-2way	35/35	0	0	0	0
test input, full runs	small-fa	35/35	0	0	0	0
test input, full runs	small-dm	35/35	0	0	0	0
test input, full runs	small-4way	35/35	0	0	0	0
test input, full runs	small-8way	35/35	0	0	0	0
train input, full runs	small-2way	31/32	1	0	0	1
train input, full runs	small-fa	31/32	1	0	0	1
train input, full runs	small-dm	31/32	1	0	0	1
train input, full runs	small-4way	31/32	1	0	0	1
train input, full runs	small-8way	31/32	1	0	0	1
ref input, SimPoint estimates	small-2way	36/36	0	0	0	0
ref input, SimPoint estimates	small-fa	36/36	0	0	0	0
ref input, SimPoint estimates	small-dm	36/36	0	0	0	0
ref input, SimPoint estimates	small-4way	36/36	0	0	0	0
ref input, SimPoint estimates	small-8way	36/36	0	0	0	0
ref input, full runs	small-2way	0/15	8	7	0	15
ref input, full runs	small-fa	0/15	8	7	0	15
ref input, full runs	small-dm	0/15	8	7	0	15
ref input, full runs	small-4way	0/15	8	7	0	15
ref input, full runs	small-8way	0/15	8	7	0	15

test input, full runs

config	completed	running	pending	other	remaining
small-2way	35/35	0	0	0	0
small-fa	35/35	0	0	0	0
small-dm	35/35	0	0	0	0
small-4way	35/35	0	0	0	0
small-8way	35/35	0	0	0	0

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	5/5	21.32%	51.31%	55.47%	21.42%	51.49%	55.80%
parest	5/5	0.22%	0.56%	1.65%	0.81%	1.48%	12.55%
deepsjeng	5/5	2.78%	9.33%	15.84%	7.31%	22.43%	46.66%
leela	5/5	0.36%	1.25%	1.66%	6.11%	15.91%	44.30%
namd	5/5	3.13%	7.91%	17.27%	3.14%	7.94%	17.37%
lbm	5/5	9.67%	27.79%	35.34%	9.67%	27.79%	35.34%
nab	5/5	7.90%	26.29%	39.86%	8.63%	27.88%	47.43%
xz	5/5	0.68%	2.35%	2.75%	3.30%	9.53%	17.13%
xz_4m_l1	5/5	0.59%	2.08%	2.43%	2.95%	8.51%	15.84%
xz_4m_l2	5/5	0.54%	1.91%	2.32%	2.73%	7.91%	15.62%
xz_4m_l3e	5/5	2.51%	9.02%	25.13%	3.76%	12.59%	42.10%
xz_4m_l4	5/5	2.12%	6.97%	14.43%	4.34%	13.23%	32.30%
xz_4m_l4e	5/5	2.28%	8.15%	25.17%	3.39%	11.30%	42.07%
xz_1m_l0	5/5	0.62%	2.50%	1.68%	3.69%	10.95%	17.49%
xz_1m_l1	5/5	0.58%	2.38%	1.48%	3.48%	10.36%	16.63%
xz_1m_l2	5/5	0.57%	2.35%	1.45%	3.39%	10.13%	16.34%
xz_1m_l3e	5/5	2.59%	9.35%	16.76%	4.60%	15.17%	33.91%
xz_1m_l4	5/5	2.01%	6.65%	9.30%	5.08%	15.11%	26.95%
xz_1m_l4e	5/5	2.43%	8.77%	16.84%	4.27%	14.07%	33.92%
imagick	5/5	0.02%	0.08%	0.05%	4.16%	11.90%	27.17%
blender	5/5	1.16%	2.72%	4.44%	9.37%	20.20%	39.50%
povray	5/5	0.27%	0.45%	0.93%	14.17%	22.21%	55.05%
omnetpp	5/5	0.09%	0.19%	0.36%	12.82%	28.51%	53.40%
xalancbmk	5/5	0.23%	0.53%	0.92%	8.96%	16.83%	45.24%
x264	5/5	6.24%	13.47%	39.69%	8.48%	18.06%	54.69%
gcc	5/5	0.84%	2.18%	3.58%	12.72%	34.38%	52.28%
mcf	5/5	1.58%	2.11%	16.47%	2.74%	3.87%	27.40%
perlbench_makerand	5/5	0.08%	0.00%	0.50%	18.63%	28.89%	59.00%
bwaves_1	5/5	3.55%	9.47%	18.87%	3.55%	9.47%	18.87%
bwaves_2	5/5	3.55%	9.47%	18.87%	3.55%	9.47%	18.87%
exchange2	5/5	5.03%	10.75%	12.58%	6.23%	13.14%	15.81%
fotonik3d	5/5	3.84%	10.81%	9.25%	3.87%	10.86%	9.40%
roms	5/5	4.79%	15.22%	9.67%	4.88%	15.37%	10.29%
wrf	5/5	0.91%	3.41%	2.49%	1.59%	4.94%	7.83%
cam4	5/5	2.11%	7.90%	7.04%	4.52%	14.20%	21.40%
pooled		5.17%	12.33%	27.44%	7.17%	16.47%	39.99%
mean		2.78%	7.88%	12.36%	6.35%	15.89%	31.03%

Miss rates: small-2way (L1D 32 sets x 2, L2 64 sets x 8)

35 of 35 runs completed, 0 running, 0 pending; **0 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.76%	4.45%	35.78%	3.91%	14.56%	64.32%	13.52%	8.96%
parest	done	11.59%	9.58%	12.36%	3.51%	2.72%	57.67%	0.55%	2.81%
deepsjeng	done	9.98%	2.86%	12.90%	7.61%	4.89%	6.89%	1.74%	13.49%
leela	done	6.90%	9.70%	7.74%	0.42%	4.44%	20.72%	0.00%	96.05%
namd	done	12.89%	5.49%	16.11%	1.31%	2.90%	25.71%	4.25%	17.28%
lbm	done	23.74%	0.42%	8.74%	2.26%	79.52%	88.75%	0.00%	3.10%
nab	done	6.91%	1.39%	10.46%	2.12%	0.91%	4.63%	0.13%	0.50%
xz	done	6.50%	0.12%	8.27%	1.00%	3.39%	25.90%	0.01%	4.24%
xz_4m_l1	done	8.00%	0.12%	9.43%	0.81%	5.52%	35.15%	0.01%	7.68%
xz_4m_l2	done	9.80%	0.12%	11.81%	0.81%	5.96%	42.07%	0.01%	8.16%
xz_4m_l3e	done	6.92%	1.91%	8.00%	1.17%	5.42%	19.19%	0.00%	1.06%
xz_4m_l4	done	8.37%	2.25%	10.00%	2.39%	5.73%	30.25%	0.00%	2.97%
xz_4m_l4e	done	6.70%	2.02%	7.58%	1.17%	5.51%	17.71%	0.00%	0.95%
xz_1m_l0	done	5.94%	0.17%	8.11%	2.01%	2.26%	29.24%	0.02%	2.93%
xz_1m_l1	done	6.73%	0.17%	8.81%	1.71%	3.19%	33.78%	0.02%	3.89%
xz_1m_l2	done	6.93%	0.15%	9.07%	1.72%	3.25%	34.23%	0.02%	3.97%
xz_1m_l3e	done	6.21%	1.54%	7.72%	2.13%	3.74%	18.66%	0.00%	0.46%
xz_1m_l4	done	6.83%	1.73%	8.77%	4.36%	3.70%	24.25%	0.00%	0.93%
xz_1m_l4e	done	6.17%	1.62%	7.52%	2.11%	3.79%	17.52%	0.00%	0.43%
imagick	done	4.42%	10.07%	3.82%	69.57%	5.76%	81.36%	35.66%	54.63%
blender	done	5.34%	2.28%	5.62%	3.16%	5.05%	19.43%	1.54%	1.25%
povray	done	10.14%	7.41%	12.18%	20.68%	4.98%	41.21%	67.06%	66.16%
omnetpp	done	12.56%	0.25%	16.96%	0.86%	4.40%	26.59%	20.83%	3.14%
xalancbmk	done	13.66%	10.00%	16.91%	18.70%	4.80%	33.74%	4.23%	4.16%
x264	done	2.33%	0.39%	2.80%	0.95%	2.40%	26.97%	1.46%	13.50%
gcc	done	7.17%	0.67%	9.36%	2.17%	4.21%	32.52%	5.35%	11.67%
mcf	done	25.50%	5.57%	28.54%	8.28%	9.83%	75.88%	6.15%	15.64%
perlbench_makerand	done	5.72%	0.00%	8.03%	2.72%	1.02%	0.00%	—	0.00%
bwaves_1	done	41.07%	2.70%	45.88%	0.33%	44.79%	31.22%	10.11%	0.24%
bwaves_2	done	41.07%	2.70%	45.89%	0.33%	44.79%	31.22%	10.11%	0.24%
exchange2	done	2.50%	2.74%	3.18%	3.56%	1.38%	1.08%	0.11%	0.44%
fotonik3d	done	17.86%	5.71%	24.31%	0.06%	6.13%	41.35%	1.22%	2.98%
roms	done	21.00%	4.44%	25.89%	1.56%	15.86%	38.82%	7.87%	9.26%
wrf	done	18.06%	9.40%	19.31%	6.61%	15.08%	22.28%	6.65%	10.05%
cam4	done	10.24%	2.66%	10.53%	2.91%	11.62%	52.99%	6.95%	31.72%
pooled		8.77%	2.14%	10.92%	1.96%	6.92%	38.69%	8.40%	10.40%
mean		11.76%	3.22%	13.95%	5.29%	9.81%	32.95%	6.05%	11.57%

Miss rates: small-fa (L1D 1 sets x 64, L2 1 sets x 512)

35 of 35 runs completed, 0 running, 0 pending; **0 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.22%	4.85%	33.94%	5.15%	12.44%	59.94%	12.41%	2.86%
parest	done	9.72%	1.35%	10.44%	1.45%	1.81%	68.60%	3.42%	9.30%
deepsjeng	done	8.08%	1.43%	10.34%	7.23%	4.24%	6.11%	2.44%	11.58%
leela	done	5.07%	0.33%	5.82%	0.42%	3.15%	27.89%	0.04%	98.92%
namd	done	11.61%	2.56%	14.73%	1.18%	2.60%	28.07%	7.56%	16.88%
lbm	done	21.23%	0.00%	7.51%	0.55%	72.77%	99.28%	—	3.25%
nab	done	3.89%	0.13%	6.20%	0.11%	0.75%	8.31%	1.60%	9.58%
xz	done	5.09%	0.01%	6.96%	0.16%	1.73%	31.78%	0.14%	25.60%
xz_4m_l1	done	6.62%	0.01%	8.12%	0.19%	3.93%	41.40%	0.14%	30.88%
xz_4m_l2	done	8.61%	0.01%	10.66%	0.21%	4.63%	47.32%	0.14%	31.02%
xz_4m_l3e	done	6.04%	0.38%	7.38%	0.30%	3.45%	21.48%	0.01%	3.78%
xz_4m_l4	done	6.39%	0.09%	8.04%	0.65%	3.69%	38.80%	0.01%	7.07%
xz_4m_l4e	done	5.90%	0.51%	7.02%	0.32%	3.53%	19.54%	0.01%	3.17%
xz_1m_l0	done	4.62%	0.01%	6.77%	0.49%	0.92%	36.56%	0.46%	11.13%
xz_1m_l1	done	5.20%	0.01%	7.25%	0.54%	1.67%	42.80%	0.46%	11.53%
xz_1m_l2	done	5.49%	0.01%	7.58%	0.55%	1.85%	42.54%	0.46%	11.61%
xz_1m_l3e	done	4.76%	0.22%	6.47%	0.27%	1.67%	23.80%	0.00%	2.67%
xz_1m_l4	done	4.64%	0.02%	6.56%	0.60%	1.64%	34.83%	0.05%	4.10%
xz_1m_l4e	done	4.81%	0.32%	6.37%	0.28%	1.72%	21.91%	0.00%	2.33%
imagick	done	4.16%	7.20%	3.49%	58.93%	5.65%	87.38%	49.88%	64.35%
blender	done	5.13%	2.12%	5.28%	3.87%	5.04%	19.66%	1.62%	1.02%
povray	done	8.70%	7.06%	10.49%	19.52%	4.14%	50.57%	71.81%	68.63%
omnetpp	done	11.53%	0.05%	15.56%	0.77%	4.05%	27.53%	23.39%	1.37%
xalancbmk	done	13.00%	10.87%	16.05%	21.93%	4.63%	34.56%	3.50%	3.19%
x264	done	1.84%	0.16%	2.22%	0.85%	1.91%	31.44%	2.43%	15.18%
gcc	done	5.69%	0.22%	7.26%	1.34%	3.62%	38.30%	15.97%	18.12%
mcf	done	24.52%	3.11%	27.48%	6.86%	9.71%	78.48%	10.04%	17.32%
perlbench_makerand	done	0.08%	0.00%	0.12%	0.26%	0.01%	0.08%	—	0.02%
bwaves_1	done	7.31%	0.86%	7.75%	0.00%	10.68%	95.60%	17.42%	36.61%
bwaves_2	done	7.31%	0.86%	7.75%	0.00%	10.68%	95.60%	17.42%	36.61%
exchange2	done	1.54%	1.69%	2.06%	2.61%	0.64%	1.79%	0.20%	0.70%
fotonik3d	done	8.89%	1.99%	11.55%	0.01%	4.82%	82.63%	1.96%	24.81%
roms	done	10.76%	2.17%	12.36%	0.97%	11.59%	64.31%	10.14%	12.82%
wrf	done	5.88%	5.43%	5.90%	4.49%	5.93%	61.76%	10.42%	19.98%
cam4	done	8.80%	1.46%	8.87%	2.47%	10.55%	59.48%	11.56%	38.18%
pooled		5.59%	1.59%	6.83%	1.94%	4.53%	52.14%	10.51%	8.37%
mean		7.66%	1.64%	9.21%	4.16%	6.34%	43.72%	8.40%	18.75%

Miss rates: small-dm (L1D 64 sets x 1, L2 512 sets x 1)

35 of 35 runs completed, 0 running, 0 pending; **0 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	17.36%	6.42%	36.76%	5.51%	16.60%	48.80%	11.27%	8.80%
parest	done	15.36%	20.70%	16.29%	6.63%	4.09%	47.58%	6.04%	4.98%
deepsjeng	done	13.69%	6.73%	17.17%	9.45%	7.46%	19.74%	5.47%	19.70%
leela	done	8.96%	19.60%	9.62%	0.46%	6.77%	17.75%	0.11%	89.76%
namd	done	15.58%	12.11%	18.89%	2.49%	3.89%	37.33%	9.19%	19.63%
lbm	done	27.22%	1.63%	13.94%	5.46%	81.64%	75.06%	5.17%	6.13%
nab	done	10.79%	1.54%	16.64%	1.65%	2.00%	15.30%	4.26%	1.90%
xz	done	8.70%	1.33%	10.47%	0.57%	5.73%	28.28%	0.77%	4.42%
xz_4m_l1	done	10.08%	1.45%	11.65%	0.59%	7.40%	36.30%	0.67%	6.24%
xz_4m_l2	done	11.65%	1.49%	13.72%	0.61%	7.75%	42.38%	0.72%	6.54%
xz_4m_l3e	done	9.40%	3.08%	10.73%	2.60%	7.62%	25.69%	2.17%	2.46%
xz_4m_l4	done	10.81%	3.07%	12.82%	3.15%	7.68%	33.53%	2.39%	6.32%
xz_4m_l4e	done	9.17%	3.19%	10.25%	2.62%	7.73%	24.56%	2.26%	2.47%
xz_1m_l0	done	7.98%	2.02%	10.31%	0.93%	4.06%	29.63%	0.74%	4.37%
xz_1m_l1	done	8.79%	2.14%	11.01%	0.94%	5.05%	33.78%	0.69%	4.81%
xz_1m_l2	done	8.92%	2.15%	11.17%	0.94%	5.09%	34.53%	0.69%	4.80%
xz_1m_l3e	done	8.68%	2.69%	10.71%	3.22%	5.26%	23.89%	1.69%	2.08%
xz_1m_l4	done	8.85%	2.90%	11.22%	4.94%	5.07%	28.51%	1.76%	3.37%
xz_1m_l4e	done	8.88%	2.79%	10.80%	3.23%	5.33%	22.40%	1.64%	2.05%
imagick	done	7.43%	16.95%	7.67%	70.35%	6.85%	47.92%	58.94%	85.67%
blender	done	9.92%	2.96%	11.28%	7.02%	7.67%	31.78%	70.70%	1.91%
povray	done	13.66%	13.93%	16.10%	21.86%	7.48%	37.82%	37.71%	55.51%
omnetpp	done	15.05%	0.72%	19.86%	10.67%	6.10%	36.64%	7.35%	0.74%
xalancbmk	done	15.16%	13.19%	18.38%	18.14%	6.36%	41.74%	10.63%	10.75%
x264	done	3.98%	0.98%	4.71%	1.20%	4.58%	26.05%	4.69%	17.50%
gcc	done	10.14%	2.27%	13.17%	4.98%	5.94%	32.44%	2.38%	7.12%
mcf	done	28.22%	8.95%	31.53%	10.17%	10.66%	72.37%	8.73%	19.17%
perlbench_makerand	done	15.27%	0.00%	19.33%	1.44%	7.06%	0.34%	—	0.01%
bwaves_1	done	44.12%	5.35%	49.10%	1.37%	47.45%	27.18%	19.22%	2.30%
bwaves_2	done	44.12%	5.35%	49.10%	1.37%	47.45%	27.18%	19.22%	2.30%
exchange2	done	4.01%	3.42%	5.09%	3.67%	2.66%	1.04%	0.14%	0.49%
fotonik3d	done	22.43%	9.89%	29.57%	0.43%	9.39%	38.96%	11.39%	2.22%
roms	done	30.01%	7.11%	37.22%	3.38%	21.21%	38.90%	15.88%	7.82%
wrf	done	26.37%	13.69%	28.28%	9.13%	21.78%	26.50%	14.64%	14.89%
cam4	done	14.02%	5.43%	14.76%	4.41%	14.73%	44.30%	11.42%	26.82%
pooled		11.58%	3.60%	14.20%	2.60%	9.46%	35.05%	9.96%	12.41%
mean		14.99%	5.92%	17.69%	6.44%	12.10%	33.03%	10.32%	13.03%

Miss rates: small-4way (L1D 16 sets x 4, L2 64 sets x 8)

35 of 35 runs completed, 0 running, 0 pending; **0 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.30%	4.85%	34.46%	4.27%	12.94%	65.69%	12.63%	8.16%
parest	done	10.21%	2.85%	10.95%	1.61%	2.08%	65.62%	1.84%	6.14%
deepsjeng	done	8.90%	2.01%	11.49%	7.25%	4.47%	7.54%	2.46%	14.06%
leela	done	5.85%	4.14%	6.65%	0.42%	3.69%	24.20%	0.00%	96.54%
namd	done	11.71%	2.78%	14.82%	1.14%	2.65%	28.28%	8.43%	19.85%
lbm	done	21.66%	0.00%	7.78%	0.68%	73.96%	97.36%	—	10.39%
nab	done	5.33%	0.51%	8.40%	0.31%	0.78%	6.02%	0.37%	3.38%
xz	done	5.69%	0.01%	7.48%	0.21%	2.53%	29.53%	0.14%	20.53%
xz_4m_l1	done	7.17%	0.01%	8.66%	0.22%	4.55%	39.21%	0.14%	27.77%
xz_4m_l2	done	9.13%	0.01%	11.23%	0.24%	5.05%	45.18%	0.14%	27.73%
xz_4m_l3e	done	6.35%	1.04%	7.55%	0.50%	4.39%	20.93%	0.00%	2.44%
xz_4m_l4	done	7.39%	0.98%	9.11%	0.99%	4.63%	34.26%	0.00%	7.15%
xz_4m_l4e	done	6.20%	1.15%	7.20%	0.52%	4.49%	19.17%	0.01%	2.14%
xz_1m_l0	done	5.14%	0.01%	7.30%	0.70%	1.47%	33.71%	0.46%	8.41%
xz_1m_l1	done	5.81%	0.01%	7.91%	0.67%	2.21%	39.07%	0.46%	9.95%
xz_1m_l2	done	6.07%	0.01%	8.24%	0.68%	2.30%	39.07%	0.46%	10.02%
xz_1m_l3e	done	5.46%	0.74%	7.12%	0.68%	2.66%	21.23%	0.00%	1.45%
xz_1m_l4	done	5.83%	0.70%	7.91%	1.36%	2.59%	28.43%	0.00%	2.97%
xz_1m_l4e	done	5.47%	0.83%	6.98%	0.68%	2.71%	19.76%	0.00%	1.33%
imagick	done	4.20%	8.47%	3.54%	68.44%	5.67%	85.57%	42.42%	55.53%
blender	done	5.06%	3.27%	5.21%	3.87%	4.93%	20.38%	1.08%	1.02%
povray	done	9.13%	6.99%	11.06%	19.61%	4.24%	45.58%	71.08%	69.40%
omnetpp	done	11.85%	0.05%	16.10%	0.82%	3.96%	28.01%	97.10%	3.28%
xalancbmk	done	13.28%	9.92%	16.46%	18.32%	4.61%	34.60%	4.26%	4.23%
x264	done	1.97%	0.22%	2.39%	0.89%	1.97%	31.00%	2.36%	14.53%
gcc	done	6.10%	0.33%	7.84%	1.48%	3.82%	36.94%	10.70%	17.33%
mcf	done	24.85%	4.00%	27.84%	7.21%	9.72%	77.84%	8.59%	17.95%
perlbench_makerand	done	3.76%	0.00%	5.10%	6.66%	1.01%	0.00%	—	0.00%
bwaves_1	done	40.95%	3.81%	45.81%	0.02%	43.46%	31.23%	6.87%	3.80%
bwaves_2	done	40.95%	3.81%	45.81%	0.02%	43.46%	31.23%	6.87%	3.80%
exchange2	done	2.01%	1.74%	2.56%	2.98%	1.15%	1.35%	0.17%	0.54%
fotonik3d	done	11.96%	3.82%	16.00%	0.01%	4.87%	61.79%	1.82%	21.07%
roms	done	14.94%	3.05%	18.09%	0.79%	12.61%	53.16%	11.50%	17.99%
wrf	done	11.55%	6.88%	12.11%	4.95%	10.35%	34.19%	9.04%	13.39%
cam4	done	9.07%	1.79%	9.16%	2.70%	10.78%	59.49%	10.34%	34.15%
pooled		7.58%	1.85%	9.44%	1.85%	5.88%	44.14%	9.85%	10.96%
mean		10.47%	2.31%	12.47%	4.63%	8.76%	37.05%	9.45%	15.95%

Miss rates: small-8way (L1D 8 sets x 8, L2 64 sets x 8)

35 of 35 runs completed, 0 running, 0 pending; **0 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.62%	5.02%	35.03%	4.67%	13.03%	64.56%	12.21%	7.59%
parest	done	9.90%	1.42%	10.63%	1.47%	1.87%	67.70%	3.69%	6.70%
deepsjeng	done	8.63%	1.77%	11.10%	7.10%	4.43%	7.70%	2.76%	14.29%
leela	done	5.16%	0.59%	5.88%	0.42%	3.30%	27.19%	0.02%	96.59%
namd	done	11.61%	2.55%	14.72%	1.17%	2.60%	28.50%	9.18%	19.45%
lbm	done	21.73%	0.00%	7.48%	0.63%	74.83%	97.05%	—	11.09%
nab	done	4.07%	0.16%	6.49%	0.13%	0.75%	7.87%	1.19%	8.18%
xz	done	5.34%	0.01%	7.14%	0.13%	2.12%	31.37%	0.14%	31.61%
xz_4m_l1	done	6.79%	0.01%	8.26%	0.17%	4.18%	41.37%	0.14%	37.15%
xz_4m_l2	done	8.81%	0.01%	10.89%	0.18%	4.77%	46.80%	0.14%	37.27%
xz_4m_l3e	done	6.09%	0.45%	7.40%	0.36%	3.71%	21.75%	0.01%	3.40%
xz_4m_l4	done	6.79%	0.23%	8.51%	0.75%	4.01%	37.27%	0.00%	9.42%
xz_4m_l4e	done	5.97%	0.57%	7.06%	0.38%	3.80%	19.86%	0.01%	2.94%
xz_1m_l0	done	4.78%	0.01%	6.93%	0.40%	1.09%	36.16%	0.46%	14.69%
xz_1m_l1	done	5.39%	0.01%	7.47%	0.44%	1.81%	42.00%	0.46%	14.97%
xz_1m_l2	done	5.68%	0.01%	7.83%	0.45%	1.95%	41.63%	0.46%	15.16%
xz_1m_l3e	done	4.97%	0.27%	6.70%	0.38%	1.90%	23.27%	0.00%	2.55%
xz_1m_l4	done	5.05%	0.09%	7.09%	0.78%	1.85%	32.77%	0.01%	5.15%
xz_1m_l4e	done	5.02%	0.37%	6.60%	0.39%	1.95%	21.50%	0.00%	2.29%
imagick	done	4.17%	7.19%	3.50%	58.77%	5.66%	86.22%	49.92%	64.67%
blender	done	5.04%	2.14%	5.15%	3.87%	5.02%	20.42%	1.64%	1.02%
povray	done	8.70%	6.93%	10.56%	19.45%	3.97%	47.73%	71.63%	69.85%
omnetpp	done	11.57%	0.05%	15.63%	0.72%	4.05%	28.59%	97.51%	3.36%
xalancbmk	done	13.12%	10.13%	16.27%	18.90%	4.51%	35.01%	4.18%	4.09%
x264	done	1.86%	0.16%	2.24%	0.86%	1.94%	32.44%	3.20%	15.00%
gcc	done	5.73%	0.23%	7.30%	1.33%	3.68%	38.80%	16.56%	19.16%
mcf	done	24.64%	3.55%	27.62%	6.92%	9.69%	78.48%	9.65%	18.71%
perlbench_makerand	done	3.80%	0.00%	5.17%	100.00%	0.51%	0.00%	—	0.00%
bwaves_1	done	26.02%	5.47%	26.51%	0.02%	43.92%	37.38%	4.30%	3.66%
bwaves_2	done	26.02%	5.47%	26.51%	0.02%	43.92%	37.38%	4.30%	3.66%
exchange2	done	1.34%	1.43%	1.75%	2.70%	0.57%	2.01%	0.21%	0.59%
fotonik3d	done	9.21%	2.50%	11.98%	0.01%	4.81%	80.25%	2.77%	29.44%
roms	done	11.83%	2.45%	13.80%	0.89%	11.92%	67.05%	14.09%	15.94%
wrf	done	6.92%	6.23%	6.91%	4.76%	7.09%	55.78%	9.85%	13.91%
cam4	done	8.84%	1.53%	8.89%	2.52%	10.63%	60.99%	12.07%	36.84%
pooled		6.44%	1.80%	7.84%	1.88%	5.44%	49.51%	10.02%	10.84%
mean		9.03%	1.97%	10.66%	6.92%	8.45%	40.14%	10.08%	18.30%

train input, full runs

config	completed	running	pending	other	remaining
small-2way	31/32	1	0	0	1
small-fa	31/32	1	0	0	1
small-dm	31/32	1	0	0	1
small-4way	31/32	1	0	0	1
small-8way	31/32	1	0	0	1

Runs not completed in every configuration:

run	small-2way	small-fa	small-dm	small-4way	small-8way
roms	running	running	running	running	running

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	5/5	21.33%	51.32%	55.49%	21.43%	51.50%	55.81%
parest	5/5	0.18%	0.45%	1.28%	0.49%	0.93%	7.38%
deepsjeng	5/5	2.99%	9.41%	15.98%	8.19%	23.80%	48.29%
leela	5/5	0.43%	1.55%	2.04%	6.08%	15.99%	46.72%
namd	5/5	3.22%	8.11%	17.74%	3.23%	8.13%	17.81%
lbm	5/5	9.69%	27.97%	35.99%	9.69%	27.97%	35.99%
nab	5/5	7.61%	25.91%	37.53%	8.29%	27.38%	44.95%
nab_gcn4dna	5/5	7.30%	24.89%	35.20%	8.25%	26.98%	45.65%
xz	5/5	1.36%	4.25%	4.82%	6.14%	16.52%	26.27%
xz_combined	5/5	2.79%	10.48%	25.95%	4.09%	14.43%	41.80%
imagick	5/5	0.04%	0.21%	4.19%	0.25%	0.82%	48.78%
blender	5/5	0.31%	0.55%	4.03%	3.18%	5.93%	39.04%
povray	5/5	0.27%	0.46%	0.95%	14.26%	22.21%	55.40%
omnetpp	5/5	0.01%	0.02%	0.03%	12.59%	24.64%	48.38%
xalancbmk	5/5	0.44%	1.00%	1.04%	18.29%	32.42%	63.05%
x264	5/5	6.30%	13.42%	40.78%	8.75%	18.41%	57.31%
gcc_200	5/5	1.16%	2.64%	4.34%	16.68%	35.31%	67.49%
gcc_scilab	5/5	1.25%	2.90%	4.70%	15.74%	33.65%	65.08%
gcc	5/5	1.41%	3.09%	5.21%	16.29%	32.84%	65.64%
mcf	5/5	1.61%	2.27%	13.28%	2.75%	4.04%	21.88%
perlbench_scrabbl	5/5	0.23%	0.30%	0.93%	17.65%	32.90%	56.14%
perlbench_suns	5/5	0.98%	2.06%	2.86%	18.85%	36.45%	59.85%
perlbench_diffmail	5/5	2.06%	4.51%	4.76%	17.54%	31.15%	53.97%
perlbench_perfect	5/5	1.44%	2.34%	5.00%	18.85%	33.31%	60.43%
perlbench_splitmail	5/5	5.79%	9.36%	20.68%	17.56%	30.55%	58.86%
bwaves_1	5/5	2.99%	7.65%	18.86%	2.99%	7.65%	18.86%
bwaves_2	5/5	3.04%	7.82%	18.83%	3.04%	7.82%	18.83%
exchange2	5/5	4.84%	10.67%	12.41%	6.03%	13.09%	15.72%
fotonik3d	5/5	3.99%	10.66%	12.65%	4.00%	10.69%	12.75%
roms	0/5	—	—	—	—	—	—
wrf	5/5	0.92%	3.44%	2.61%	1.59%	4.97%	8.09%

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cam4	5/5	2.49%	9.15%	7.96%	4.59%	14.65%	19.40%
pooled		3.60%	9.44%	17.30%	7.34%	17.15%	41.91%
mean		3.18%	8.35%	13.49%	9.59%	20.87%	41.47%

Miss rates: small-2way (L1D 32 sets x 2, L2 64 sets x 8)

31 of 32 runs completed, 1 running, 0 pending; **1 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.77%	4.46%	35.81%	3.90%	14.55%	64.32%	13.50%	8.93%
parest	done	11.51%	4.71%	12.25%	2.95%	2.54%	61.90%	1.42%	7.39%
deepsjeng	done	9.80%	2.94%	12.68%	7.87%	4.93%	6.58%	1.66%	12.65%
leela	done	6.41%	7.02%	7.20%	0.20%	4.07%	16.65%	0.00%	85.10%
namd	done	12.79%	5.33%	16.02%	1.39%	2.74%	25.42%	4.25%	16.19%
lbm	done	23.36%	0.42%	8.67%	2.11%	79.55%	88.88%	0.00%	7.03%
nab	done	7.24%	1.00%	11.16%	0.81%	0.72%	4.20%	0.07%	0.42%
nab_gcn4dna	done	6.93%	1.14%	10.44%	1.38%	0.69%	41.67%	1.38%	0.64%
xz	done	7.11%	1.41%	9.74%	9.93%	2.79%	32.18%	0.00%	0.93%
xz_combined	done	10.85%	3.70%	13.43%	2.30%	5.39%	20.84%	0.02%	0.77%
imagick	done	5.02%	34.99%	5.00%	0.47%	1.97%	94.86%	1.01%	4.96%
blender	done	7.42%	3.84%	8.01%	7.01%	3.57%	11.64%	2.32%	6.49%
povray	done	10.10%	7.63%	12.08%	20.37%	5.06%	41.87%	62.02%	68.89%
omnetpp	done	16.55%	0.25%	22.02%	0.87%	5.81%	28.83%	20.84%	2.89%
xalancbmk	done	12.40%	20.48%	16.05%	29.68%	4.86%	12.36%	0.00%	0.04%
x264	done	2.33%	0.39%	2.83%	0.94%	2.27%	28.09%	1.59%	13.97%
gcc_200	done	9.56%	2.26%	12.77%	5.11%	3.83%	36.44%	2.29%	7.98%
gcc_scilab	done	10.17%	2.62%	13.60%	4.68%	4.02%	36.72%	1.61%	8.78%
gcc	done	9.94%	2.41%	13.21%	4.85%	3.93%	38.95%	1.98%	8.43%
mcf	done	23.82%	7.04%	27.65%	8.49%	7.41%	75.08%	4.48%	16.24%
perlbench_scrabbl	done	7.97%	3.14%	10.83%	11.63%	3.08%	11.62%	0.00%	0.01%
perlbench_suns	done	6.55%	1.11%	9.10%	9.00%	2.40%	18.32%	0.09%	12.14%
perlbench_diffmail	done	6.46%	4.87%	8.65%	4.17%	2.66%	32.33%	0.01%	14.46%
perlbench_perfect	done	9.27%	1.24%	12.65%	5.86%	3.58%	51.16%	21.32%	24.44%
perlbench_splitmail	done	5.38%	1.79%	7.57%	2.78%	2.31%	17.25%	0.04%	3.12%
bwaves_1	done	42.24%	3.28%	46.73%	0.27%	40.60%	27.05%	5.37%	0.17%
bwaves_2	done	42.11%	3.21%	46.64%	0.27%	41.05%	27.48%	5.43%	0.17%
exchange2	done	2.26%	2.05%	2.85%	2.70%	1.43%	1.44%	0.18%	0.74%
fotonik3d	done	17.95%	5.70%	24.13%	0.06%	1.99%	38.50%	1.27%	2.89%
roms	running	—	—	—	—	—	—	—	—
wrf	done	17.59%	9.17%	18.66%	6.10%	15.15%	22.33%	6.47%	9.56%
cam4	done	11.20%	2.59%	11.85%	2.62%	12.38%	54.29%	7.88%	25.84%
pooled		11.76%	3.44%	14.32%	3.36%	7.59%	36.78%	8.06%	9.01%
mean		12.52%	4.91%	15.17%	5.19%	9.27%	34.49%	5.44%	12.01%

Miss rates: small-fa (L1D 1 sets x 64, L2 1 sets x 512)

31 of 32 runs completed, 1 running, 0 pending; **1 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.23%	4.85%	33.96%	5.15%	12.45%	59.95%	12.41%	2.86%
parest	done	9.87%	1.06%	10.56%	1.25%	1.68%	72.07%	5.55%	14.15%
deepsjeng	done	7.79%	1.45%	9.88%	7.51%	4.35%	6.05%	2.15%	10.50%
leela	done	4.49%	0.20%	5.23%	0.20%	2.49%	22.42%	0.04%	98.69%
namd	done	11.49%	2.42%	14.61%	1.26%	2.43%	27.80%	7.69%	15.51%
lbm	done	20.86%	0.00%	7.41%	0.37%	72.77%	99.62%	—	29.21%
nab	done	3.87%	0.09%	6.11%	0.17%	0.57%	7.20%	0.52%	1.97%
nab_gcn4dna	done	3.91%	0.06%	6.13%	0.21%	0.46%	74.36%	24.66%	4.31%
xz	done	4.77%	0.03%	7.48%	0.70%	0.57%	47.29%	0.00%	10.38%
xz_combined	done	9.79%	0.84%	12.72%	0.51%	3.88%	22.72%	0.01%	2.94%
imagick	done	5.01%	34.95%	5.00%	0.40%	1.76%	94.98%	1.01%	6.14%
blender	done	6.70%	4.77%	7.32%	2.35%	2.78%	11.68%	1.99%	18.36%
povray	done	8.60%	7.07%	10.48%	19.32%	3.80%	51.48%	71.97%	68.16%
omnetpp	done	16.03%	0.05%	21.36%	0.79%	5.55%	28.39%	23.39%	1.45%
xalancbmk	done	11.44%	6.74%	14.75%	32.17%	4.80%	6.67%	0.00%	0.01%
x264	done	1.80%	0.15%	2.20%	0.83%	1.81%	33.73%	2.63%	16.31%
gcc_200	done	7.53%	0.81%	10.03%	4.07%	3.14%	43.03%	4.22%	7.80%
gcc_scilab	done	8.13%	1.01%	10.86%	3.83%	3.32%	42.90%	2.93%	9.17%
gcc	done	7.80%	0.74%	10.38%	4.11%	3.12%	46.27%	5.37%	7.98%
mcf	done	22.96%	3.67%	26.70%	7.07%	7.37%	77.43%	7.74%	17.84%
perlbench_scrabbl	done	4.94%	5.08%	7.05%	5.74%	1.33%	6.46%	0.00%	0.02%
perlbench_suns	done	4.62%	0.13%	6.60%	3.31%	1.50%	20.06%	0.67%	11.88%
perlbench_diffmail	done	3.03%	0.15%	4.27%	2.24%	1.03%	53.90%	0.03%	24.65%
perlbench_perfect	done	7.52%	0.45%	10.69%	5.10%	2.11%	56.72%	58.32%	31.08%
perlbench_splitmail	done	3.49%	0.05%	4.99%	1.87%	1.56%	20.52%	1.47%	2.58%
bwaves_1	done	7.29%	0.74%	7.71%	0.00%	10.06%	97.31%	14.74%	98.73%
bwaves_2	done	7.29%	0.74%	7.71%	0.00%	10.13%	97.23%	14.39%	98.71%
exchange2	done	1.32%	1.29%	1.84%	1.88%	0.52%	2.51%	0.31%	1.23%
fotonik3d	done	8.39%	1.99%	11.57%	0.01%	0.12%	82.20%	2.28%	24.30%
roms	running	—	—	—	—	—	—	—	—
wrf	done	5.78%	5.29%	5.76%	4.18%	5.97%	61.40%	10.19%	18.33%
cam4	done	9.81%	1.48%	10.17%	1.85%	11.47%	60.40%	12.18%	37.25%
pooled		8.18%	2.52%	9.85%	3.36%	5.42%	46.91%	10.69%	6.05%
mean		8.11%	2.85%	10.05%	3.82%	5.96%	46.28%	9.63%	22.34%

Miss rates: small-dm (L1D 64 sets x 1, L2 512 sets x 1)

31 of 32 runs completed, 1 running, 0 pending; **1 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	17.36%	6.42%	36.79%	5.50%	16.60%	48.81%	11.44%	8.83%
parest	done	15.29%	13.83%	16.21%	5.54%	3.80%	50.52%	24.84%	9.30%
deepsjeng	done	13.57%	6.64%	17.09%	9.69%	7.53%	20.58%	4.92%	19.04%
leela	done	7.77%	12.09%	8.63%	0.23%	5.06%	15.93%	0.08%	54.80%
namd	done	15.53%	11.95%	18.84%	2.50%	3.73%	37.27%	8.80%	21.24%
lbm	done	26.83%	1.63%	13.80%	5.34%	81.66%	74.99%	5.20%	5.54%
nab	done	11.61%	4.17%	16.71%	3.10%	1.82%	16.55%	1.10%	1.00%
nab_gcn4dna	done	11.34%	4.33%	16.03%	3.99%	1.54%	31.28%	3.87%	1.80%
xz	done	8.85%	3.59%	11.81%	9.75%	4.03%	32.06%	1.52%	2.69%
xz_combined	done	14.68%	7.09%	17.42%	5.23%	8.97%	25.02%	1.95%	1.77%
imagick	done	5.62%	28.95%	5.61%	2.48%	3.03%	86.53%	17.31%	2.92%
blender	done	9.96%	6.67%	10.51%	7.11%	6.45%	27.63%	13.01%	14.24%
povray	done	14.30%	12.02%	16.75%	21.55%	8.08%	36.10%	32.70%	49.08%
omnetpp	done	18.54%	0.72%	24.19%	10.69%	7.45%	41.67%	7.36%	0.74%
xalancbmk	done	14.58%	30.23%	18.61%	25.47%	6.18%	26.71%	27.23%	43.28%
x264	done	4.04%	0.93%	4.85%	1.20%	4.46%	26.31%	3.81%	16.05%
gcc_200	done	12.95%	7.30%	16.90%	8.80%	5.66%	37.34%	3.02%	9.41%
gcc_scilab	done	13.63%	7.84%	17.82%	8.20%	5.88%	38.00%	4.27%	10.91%
gcc	done	13.47%	6.70%	17.48%	8.24%	5.86%	39.11%	2.99%	9.24%
mcf	done	26.68%	10.49%	30.85%	10.54%	8.39%	71.66%	9.97%	20.62%
perlbench_scrabbl	done	12.39%	5.42%	15.67%	15.50%	6.79%	16.92%	0.00%	1.00%
perlbench_suns	done	9.86%	2.95%	12.95%	10.61%	4.92%	25.17%	2.61%	13.53%
perlbench_diffmail	done	10.33%	2.39%	13.31%	10.54%	5.51%	26.66%	0.94%	10.19%
perlbench_perfect	done	11.06%	1.54%	14.81%	7.69%	4.77%	49.94%	19.62%	12.13%
perlbench_splitmail	done	8.89%	3.08%	12.28%	2.97%	4.68%	21.52%	0.16%	7.66%
bwaves_1	done	43.95%	6.18%	48.39%	1.30%	42.07%	26.39%	19.27%	1.31%
bwaves_2	done	43.91%	6.08%	48.40%	1.32%	42.55%	26.65%	19.10%	1.31%
exchange2	done	3.65%	2.78%	4.62%	2.44%	2.62%	1.48%	0.21%	0.90%
fotonik3d	done	22.47%	9.88%	28.82%	0.43%	6.62%	36.82%	11.33%	2.27%
roms	running	—	—	—	—	—	—	—	—
wrf	done	25.82%	13.72%	27.42%	8.46%	22.13%	26.35%	14.83%	15.87%
cam4	done	14.71%	5.03%	15.94%	3.66%	15.14%	46.92%	11.55%	24.53%
pooled		14.84%	5.98%	17.76%	4.80%	9.95%	36.07%	10.04%	11.39%
mean		15.60%	7.83%	18.69%	7.10%	11.42%	35.13%	9.19%	12.68%

Miss rates: small-4way (L1D 16 sets x 4, L2 64 sets x 8)

31 of 32 runs completed, 1 running, 0 pending; **1 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.30%	4.85%	34.48%	4.27%	12.94%	65.68%	12.63%	8.09%
parest	done	10.27%	2.03%	10.97%	1.46%	1.95%	69.50%	3.10%	14.83%
deepsjeng	done	8.71%	2.02%	11.23%	7.46%	4.53%	7.25%	2.38%	13.25%
leela	done	5.26%	1.74%	5.99%	0.20%	3.26%	20.07%	0.00%	85.89%
namd	done	11.59%	2.64%	14.71%	1.23%	2.49%	28.04%	8.63%	18.29%
lbm	done	21.30%	0.00%	7.69%	0.50%	73.99%	97.61%	—	29.39%
nab	done	5.48%	0.42%	8.59%	0.27%	0.59%	5.51%	0.15%	1.23%
nab_gcn4dna	done	4.91%	0.24%	7.65%	0.35%	0.51%	58.63%	6.67%	2.57%
xz	done	6.00%	0.87%	8.78%	2.99%	1.65%	38.14%	0.00%	3.07%
xz_combined	done	9.82%	1.75%	12.52%	0.81%	4.50%	23.02%	0.04%	2.16%
imagick	done	5.01%	34.95%	5.00%	0.48%	1.76%	94.98%	1.01%	4.90%
blender	done	6.71%	5.33%	7.30%	4.78%	2.85%	12.80%	1.66%	9.49%
povray	done	9.02%	6.99%	10.95%	19.46%	4.08%	46.64%	67.66%	72.00%
omnetpp	done	15.94%	0.05%	21.32%	0.83%	5.39%	29.81%	97.14%	3.02%
xalancbmk	done	11.88%	15.33%	15.40%	26.56%	4.71%	12.81%	0.00%	0.04%
x264	done	1.97%	0.22%	2.41%	0.88%	1.87%	32.24%	2.56%	15.02%
gcc_200	done	8.32%	1.20%	11.14%	4.23%	3.34%	41.08%	4.22%	9.44%
gcc_scilab	done	8.90%	1.48%	11.93%	3.91%	3.52%	41.20%	2.79%	10.33%
gcc	done	8.69%	1.46%	11.60%	4.12%	3.38%	43.73%	3.52%	9.80%
mcf	done	23.17%	5.26%	26.93%	7.39%	7.33%	77.15%	6.07%	18.62%
perlbench_scrabbl	done	6.20%	3.29%	8.86%	8.91%	1.65%	14.10%	0.00%	0.01%
perlbench_suns	done	5.44%	0.39%	7.71%	8.94%	1.73%	20.70%	0.26%	12.21%
perlbench_diffmail	done	4.67%	0.16%	6.45%	2.76%	1.85%	41.21%	0.21%	22.43%
perlbench_perfect	done	8.25%	0.67%	11.52%	5.52%	2.71%	55.17%	39.20%	25.84%
perlbench_splitmail	done	4.50%	0.17%	6.51%	2.96%	1.69%	19.98%	0.42%	2.44%
bwaves_1	done	43.00%	4.98%	47.49%	0.14%	40.88%	26.58%	3.92%	0.32%
bwaves_2	done	42.86%	4.85%	47.39%	0.14%	41.29%	27.02%	3.99%	0.32%
exchange2	done	1.85%	1.40%	2.44%	2.35%	1.02%	1.76%	0.26%	0.87%
fotonik3d	done	11.67%	3.81%	15.96%	0.01%	0.20%	59.27%	1.90%	20.68%
roms	running	—	—	—	—	—	—	—	—
wrf	done	11.05%	6.76%	11.47%	4.70%	10.29%	34.94%	8.73%	12.40%
cam4	done	10.12%	1.77%	10.55%	2.13%	11.66%	59.86%	11.58%	31.83%
pooled		10.27%	2.96%	12.53%	3.12%	6.51%	41.62%	9.53%	9.65%
mean		11.22%	3.78%	13.64%	4.22%	8.38%	38.92%	9.69%	14.86%

Miss rates: small-8way (L1D 8 sets x 8, L2 64 sets x 8)

31 of 32 runs completed, 1 running, 0 pending; **1 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.63%	5.02%	35.06%	4.67%	13.03%	64.55%	12.21%	7.58%
parest	done	10.03%	1.20%	10.73%	1.24%	1.75%	71.14%	5.24%	16.99%
deepsjeng	done	8.41%	1.80%	10.80%	7.31%	4.48%	7.44%	2.65%	13.44%
leela	done	4.79%	0.25%	5.52%	0.20%	2.81%	21.89%	0.03%	85.88%
namd	done	11.48%	2.41%	14.60%	1.25%	2.44%	28.27%	9.44%	17.93%
lbm	done	21.37%	0.00%	7.38%	0.46%	74.88%	97.30%	—	32.18%
nab	done	4.04%	0.11%	6.39%	0.18%	0.56%	7.34%	0.55%	1.90%
nab_gcn4dna	done	4.04%	0.08%	6.31%	0.27%	0.49%	71.26%	20.08%	3.32%
xz	done	5.01%	0.08%	7.72%	1.39%	0.80%	45.61%	0.00%	6.56%
xz_combined	done	9.70%	1.02%	12.52%	0.74%	4.12%	23.25%	0.05%	2.37%
imagick	done	5.01%	34.95%	5.00%	0.40%	1.76%	94.98%	1.01%	5.81%
blender	done	6.67%	4.89%	7.28%	4.40%	2.74%	12.85%	1.81%	10.36%
povray	done	8.58%	6.97%	10.50%	19.27%	3.66%	48.92%	67.90%	72.66%
omnetpp	done	15.90%	0.05%	21.21%	0.73%	5.49%	29.84%	97.51%	3.06%
xalancbmk	done	11.40%	12.13%	14.65%	23.99%	4.81%	13.19%	0.00%	0.04%
x264	done	1.85%	0.15%	2.26%	0.85%	1.84%	33.89%	3.40%	15.49%
gcc_200	done	7.83%	0.90%	10.45%	4.05%	3.21%	43.28%	5.65%	9.85%
gcc_scilab	done	8.43%	1.15%	11.28%	3.78%	3.39%	43.14%	3.62%	10.75%
gcc	done	8.17%	0.98%	10.89%	4.01%	3.21%	46.10%	5.28%	10.13%
mcf	done	22.97%	4.49%	26.71%	7.12%	7.31%	77.82%	6.96%	19.34%
perlbench_scrabbl	done	5.68%	4.63%	8.10%	9.31%	1.52%	14.81%	0.00%	0.01%
perlbench_suns	done	4.98%	0.17%	7.12%	5.35%	1.57%	22.02%	0.60%	20.40%
perlbench_diffmail	done	4.49%	0.13%	6.21%	5.79%	1.61%	42.24%	0.27%	10.70%
perlbench_perfect	done	7.71%	0.63%	10.89%	5.22%	2.31%	57.56%	41.87%	27.88%
perlbench_splitmail	done	4.02%	0.04%	5.90%	2.07%	1.50%	21.83%	1.89%	3.31%
bwaves_1	done	24.89%	4.88%	25.55%	0.09%	39.60%	36.09%	3.57%	0.50%
bwaves_2	done	24.95%	4.75%	25.60%	0.09%	40.03%	36.24%	3.63%	0.49%
exchange2	done	1.08%	1.02%	1.45%	2.04%	0.45%	2.98%	0.36%	1.01%
fotonik3d	done	8.73%	2.49%	11.99%	0.01%	0.13%	79.16%	2.89%	29.04%
roms	running	—	—	—	—	—	—	—	—
wrf	done	6.62%	6.01%	6.56%	4.45%	6.97%	56.91%	9.74%	13.09%
cam4	done	9.86%	1.52%	10.23%	1.90%	11.53%	61.38%	13.45%	35.85%
pooled		8.99%	2.81%	10.81%	3.21%	6.11%	45.97%	10.00%	9.47%
mean		9.49%	3.38%	11.51%	3.96%	8.06%	42.36%	10.72%	15.74%

ref input, SimPoint estimates

config	completed	running	pending	other	remaining
small-2way	36/36	0	0	0	0
small-fa	36/36	0	0	0	0
small-dm	36/36	0	0	0	0
small-4way	36/36	0	0	0	0
small-8way	36/36	0	0	0	0

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	5/5	20.65%	47.95%	56.30%	20.67%	48.00%	56.39%
parest	5/5	0.16%	0.43%	0.62%	0.64%	1.19%	7.86%
deepsjeng	5/5	2.92%	9.17%	16.25%	8.11%	23.58%	49.72%
leela	5/5	0.38%	1.36%	1.80%	6.07%	15.90%	45.53%
namd	5/5	3.27%	8.21%	17.87%	3.27%	8.22%	17.90%
lbm	5/5	10.48%	32.28%	44.97%	10.48%	32.28%	44.97%
nab	5/5	5.20%	15.37%	34.02%	6.75%	18.74%	49.89%
xz_cld	5/5	3.38%	12.11%	22.88%	5.08%	17.35%	36.71%
xz_cpu2006docs	5/5	1.47%	5.78%	23.30%	2.03%	7.38%	37.90%
xz_combined	5/5	2.85%	10.66%	26.32%	4.17%	14.65%	42.32%
imagick	5/5	0.23%	0.12%	3.44%	5.33%	12.37%	46.11%
blender	5/5	0.38%	0.88%	3.23%	3.72%	6.96%	42.51%
povray	5/5	0.27%	0.46%	0.95%	14.26%	22.18%	55.24%
omnetpp	5/5	0.00%	0.00%	0.00%	12.75%	24.67%	48.13%
xalancbmk	5/5	0.13%	0.26%	0.80%	4.92%	8.32%	39.84%
x264_pass1	5/5	7.07%	15.02%	38.53%	10.18%	21.51%	55.90%
x264_pass2	5/5	6.25%	13.38%	40.16%	8.63%	18.23%	56.21%
x264_seek	5/5	6.41%	13.66%	41.07%	8.82%	18.56%	57.26%
gcc_pp_O3	5/5	1.23%	2.73%	4.48%	16.24%	33.66%	64.23%
gcc_pp_O2	5/5	1.25%	2.75%	4.47%	16.60%	34.40%	63.73%
gcc_smaller	5/5	1.09%	2.19%	4.61%	13.76%	25.29%	64.09%
gcc_ref32_O5	5/5	0.91%	1.84%	3.40%	15.53%	30.63%	59.73%
gcc_ref32_O3	5/5	0.68%	1.85%	1.06%	11.25%	27.66%	19.94%
mcf	5/5	1.40%	2.06%	7.74%	2.33%	3.60%	12.43%
perlbench_checkspam	5/5	7.98%	9.83%	29.42%	22.33%	36.24%	69.19%
perlbench_diffmail	5/5	1.96%	4.19%	4.64%	17.81%	31.25%	54.38%
perlbench_splitmail	5/5	4.16%	4.19%	15.82%	23.27%	35.82%	68.04%
bwaves_1	5/5	2.51%	6.24%	18.43%	2.51%	6.24%	18.43%
bwaves_2	5/5	2.59%	6.43%	18.99%	2.59%	6.43%	18.99%
bwaves_3	5/5	2.51%	6.23%	18.45%	2.51%	6.23%	18.45%
bwaves_4	5/5	2.56%	6.32%	19.44%	2.56%	6.32%	19.44%
exchange2	5/5	5.23%	10.55%	12.68%	6.08%	12.17%	14.84%
fotonik3d	5/5	3.36%	9.32%	9.25%	3.38%	9.37%	9.40%
roms	5/5	4.37%	13.83%	8.97%	4.44%	13.96%	9.48%
wrf	5/5	0.91%	3.38%	2.59%	1.52%	4.76%	7.57%
cam4	5/5	3.14%	11.02%	11.03%	4.99%	15.77%	21.23%
pooled		3.78%	9.06%	17.53%	7.49%	16.44%	39.26%

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
mean		3.31%	8.11%	15.78%	8.49%	18.33%	39.00%

Miss rates: small-2way (L1D 32 sets x 2, L2 64 sets x 8)

36 of 36 runs completed, 0 running, 0 pending; **0 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.64%	4.39%	33.97%	3.60%	14.03%	64.77%	12.85%	11.23%
parest	done	11.55%	3.60%	12.23%	3.31%	5.40%	63.65%	2.05%	13.78%
deepsjeng	done	9.86%	2.97%	12.82%	7.98%	4.55%	5.80%	1.57%	13.30%
leela	done	6.48%	4.29%	7.20%	0.04%	4.50%	18.20%	0.00%	0.77%
namd	done	12.51%	3.31%	15.86%	1.36%	2.70%	25.47%	5.50%	14.95%
lbm	done	19.64%	0.44%	7.67%	1.75%	81.24%	86.89%	0.00%	0.80%
nab	done	7.99%	1.57%	10.82%	2.82%	0.59%	40.00%	0.77%	0.40%
xz_cld	done	11.84%	3.97%	16.53%	1.39%	4.08%	31.57%	0.01%	0.28%
xz_cpu2006docs	done	9.62%	2.83%	10.47%	2.10%	7.10%	38.70%	3.73%	6.05%
xz_combined	done	10.79%	3.98%	13.39%	1.10%	5.56%	22.05%	0.00%	1.53%
imagick	done	2.17%	3.94%	2.76%	0.03%	0.13%	96.75%	57.27%	20.17%
blender	done	8.22%	2.76%	8.95%	2.06%	3.91%	17.84%	1.16%	32.07%
povray	done	10.32%	7.39%	12.74%	19.42%	4.20%	40.97%	50.50%	60.19%
omnetpp	done	16.65%	0.09%	22.25%	2.02%	5.74%	27.55%	45.06%	1.97%
xalancbmk	done	17.45%	10.52%	20.09%	15.82%	4.71%	68.78%	6.85%	7.90%
x264_pass1	done	2.78%	0.23%	3.36%	0.93%	3.47%	25.29%	3.67%	16.74%
x264_pass2	done	2.28%	0.36%	2.75%	0.87%	2.44%	31.33%	0.45%	16.85%
x264_seek	done	2.30%	0.32%	2.81%	0.97%	2.31%	27.65%	0.54%	12.70%
gcc_pp_O3	done	9.96%	2.05%	13.24%	5.06%	4.18%	37.45%	3.68%	11.87%
gcc_pp_O2	done	10.05%	2.16%	13.34%	5.08%	4.44%	34.58%	3.71%	10.26%
gcc_smaller	done	11.47%	2.78%	14.61%	4.43%	4.02%	43.86%	2.65%	8.57%
gcc_ref32_O5	done	10.16%	3.21%	13.28%	5.14%	4.40%	44.91%	3.95%	10.21%
gcc_ref32_O3	done	11.32%	3.92%	13.63%	6.15%	9.81%	68.76%	2.73%	15.62%
mcf	done	22.05%	7.93%	28.09%	7.61%	4.73%	72.44%	2.72%	16.32%
perlbench_checkspam	done	4.54%	0.85%	7.04%	1.78%	1.66%	16.93%	0.31%	2.13%
perlbench_diffmail	done	6.04%	2.43%	7.82%	7.79%	2.97%	30.54%	0.04%	6.97%
perlbench_splitmail	done	9.96%	0.40%	14.40%	4.28%	3.49%	7.65%	0.22%	0.74%
bwaves_1	done	43.27%	3.94%	47.38%	0.20%	36.83%	24.07%	4.55%	0.30%
bwaves_2	done	42.95%	3.80%	47.10%	0.23%	37.20%	24.39%	4.24%	0.35%
bwaves_3	done	42.88%	4.74%	46.93%	0.17%	36.28%	24.27%	4.24%	0.40%
bwaves_4	done	42.81%	3.89%	46.86%	0.16%	37.10%	24.15%	3.57%	0.27%
exchange2	done	1.83%	1.62%	2.23%	2.78%	1.19%	0.85%	0.08%	0.32%
fotonik3d	done	16.67%	6.14%	22.90%	0.17%	1.24%	41.15%	1.27%	1.81%
roms	done	21.55%	4.01%	26.62%	1.25%	15.16%	38.82%	7.76%	5.84%
wrf	done	17.88%	9.00%	18.88%	5.95%	15.69%	21.73%	6.33%	8.69%
cam4	done	11.80%	3.18%	12.89%	3.08%	12.83%	50.53%	5.20%	19.51%
pooled		12.40%	3.33%	15.42%	2.81%	7.56%	41.46%	8.56%	11.14%
mean		14.31%	3.42%	17.05%	3.58%	10.83%	37.23%	6.92%	9.77%

Miss rates: small-fa (L1D 1 sets x 64, L2 1 sets x 512)

36 of 36 runs completed, 0 running, 0 pending; **0 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.37%	4.99%	32.48%	4.97%	12.55%	61.01%	12.41%	4.79%
parest	done	9.78%	1.18%	10.35%	1.28%	4.82%	75.05%	4.89%	23.16%
deepsjeng	done	7.89%	1.59%	10.13%	7.48%	3.94%	5.04%	2.01%	11.57%
leela	done	4.02%	0.06%	4.59%	0.00%	2.53%	28.13%	0.40%	20.61%
namd	done	11.54%	2.41%	14.70%	1.29%	2.41%	27.21%	7.40%	15.85%
lbm	done	17.10%	0.00%	6.08%	0.02%	73.68%	99.98%	100.00%	54.85%
nab	done	3.91%	0.08%	5.49%	0.25%	0.42%	82.17%	14.86%	4.12%
xz_cld	done	10.09%	0.50%	15.00%	0.27%	2.20%	36.73%	0.10%	1.48%
xz_cpu2006docs	done	9.30%	1.37%	10.33%	0.97%	5.74%	39.57%	7.56%	12.89%
xz_combined	done	9.69%	0.76%	12.63%	0.47%	3.86%	24.09%	0.00%	3.30%
imagick	done	2.22%	3.94%	2.83%	0.01%	0.12%	100.00%	57.17%	54.03%
blender	done	8.01%	2.59%	8.78%	1.35%	3.42%	17.55%	1.23%	48.89%
povray	done	8.49%	7.25%	10.56%	18.47%	3.20%	50.78%	70.37%	67.22%
omnetpp	done	16.24%	0.08%	21.68%	1.36%	5.63%	27.41%	15.97%	2.17%
xalancbmk	done	16.81%	11.16%	19.36%	22.22%	4.43%	70.85%	7.06%	6.03%
x264_pass1	done	2.29%	0.14%	2.69%	1.01%	3.06%	24.98%	3.70%	14.18%
x264_pass2	done	1.83%	0.11%	2.22%	0.71%	1.96%	36.21%	1.16%	20.88%
x264_seek	done	1.83%	0.12%	2.23%	0.82%	1.89%	32.07%	1.21%	14.73%
gcc_pp_O3	done	8.03%	0.78%	10.65%	4.61%	3.42%	43.50%	8.19%	10.94%
gcc_pp_O2	done	8.27%	0.77%	10.98%	4.65%	3.68%	39.71%	6.79%	9.19%
gcc_smaller	done	9.91%	1.35%	12.74%	3.23%	3.27%	51.24%	4.10%	9.62%
gcc_ref32_O5	done	7.92%	0.97%	10.34%	3.45%	3.54%	54.04%	11.27%	11.98%
gcc_ref32_O3	done	10.27%	1.78%	11.48%	5.26%	9.56%	74.31%	3.44%	15.00%
mcf	done	21.77%	2.59%	27.84%	6.13%	4.77%	72.93%	6.79%	18.85%
perlbench_checkspam	done	1.73%	0.09%	2.73%	0.31%	0.74%	35.71%	3.13%	12.02%
perlbench_diffmail	done	2.75%	0.17%	3.88%	2.31%	0.91%	51.55%	0.00%	21.79%
perlbench_splitmail	done	10.14%	0.03%	14.00%	10.37%	3.67%	6.97%	3.07%	0.24%
bwaves_1	done	7.33%	0.75%	7.74%	0.00%	9.36%	97.66%	18.52%	98.01%
bwaves_2	done	7.29%	0.74%	7.70%	0.00%	9.42%	97.59%	17.34%	99.48%
bwaves_3	done	7.30%	0.81%	7.70%	0.00%	9.35%	97.86%	20.74%	98.31%
bwaves_4	done	7.25%	0.58%	7.67%	0.00%	9.37%	98.43%	14.15%	53.65%
exchange2	done	1.13%	1.29%	1.57%	1.65%	0.43%	1.41%	0.12%	0.63%
fotonik3d	done	8.23%	2.06%	11.58%	0.01%	0.02%	83.12%	2.44%	18.37%
roms	done	10.86%	1.55%	12.52%	0.72%	11.26%	67.88%	13.91%	6.85%
wrf	done	5.68%	5.10%	5.68%	3.73%	5.79%	62.89%	10.21%	17.07%
cam4	done	10.39%	1.80%	11.18%	2.11%	12.00%	56.58%	7.50%	25.29%
pooled		8.19%	2.52%	9.92%	2.88%	5.53%	57.40%	11.76%	8.58%
mean		8.41%	1.71%	10.28%	3.10%	6.57%	53.67%	12.76%	25.22%

Miss rates: small-dm (L1D 64 sets x 1, L2 512 sets x 1)

36 of 36 runs completed, 0 running, 0 pending; **0 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	16.94%	6.47%	34.17%	5.01%	16.06%	55.00%	19.96%	15.81%
parest	done	15.19%	10.39%	16.13%	5.80%	6.57%	51.66%	4.51%	9.35%
deepsjeng	done	13.61%	6.44%	17.15%	9.67%	7.29%	14.36%	6.53%	23.10%
leela	done	7.98%	7.04%	8.76%	0.28%	5.79%	17.91%	0.14%	0.36%
namd	done	14.97%	6.99%	18.59%	2.52%	3.62%	32.32%	9.33%	14.52%
lbm	done	22.89%	1.68%	12.45%	5.04%	83.18%	78.54%	3.75%	1.81%
nab	done	13.74%	3.93%	18.49%	2.85%	1.41%	31.34%	1.48%	3.09%
xz_cld	done	15.16%	3.62%	20.80%	2.69%	6.66%	34.07%	3.06%	1.80%
xz_cpu2006docs	done	11.35%	4.71%	12.12%	3.05%	9.83%	41.33%	5.77%	7.82%
xz_combined	done	14.39%	5.20%	17.33%	3.88%	9.21%	28.18%	2.29%	2.00%
imagick	done	2.76%	6.34%	3.43%	1.40%	0.40%	68.78%	40.68%	0.61%
blender	done	10.33%	7.50%	10.97%	3.36%	6.54%	31.65%	4.85%	21.64%
povray	done	14.40%	9.21%	17.67%	22.20%	6.19%	37.67%	32.76%	47.94%
omnetpp	done	19.33%	12.85%	25.22%	2.13%	7.83%	37.69%	0.05%	4.21%
xalancbmk	done	18.92%	13.80%	21.61%	17.84%	5.88%	68.27%	8.55%	24.98%
x264_pass1	done	4.66%	0.69%	5.80%	1.38%	5.07%	24.17%	3.30%	13.63%
x264_pass2	done	4.16%	0.78%	5.00%	1.11%	4.67%	28.76%	4.93%	17.35%
x264_seek	done	4.25%	0.92%	5.09%	1.37%	4.76%	23.46%	3.09%	13.24%
gcc_pp_O3	done	13.49%	7.07%	17.49%	8.26%	6.21%	38.93%	3.57%	15.30%
gcc_pp_O2	done	13.50%	6.93%	17.51%	8.63%	6.45%	37.38%	5.64%	13.67%
gcc_smaller	done	14.57%	6.22%	18.06%	7.17%	6.18%	45.12%	7.65%	10.35%
gcc_ref32_O5	done	13.63%	7.02%	17.52%	8.71%	6.31%	44.76%	6.86%	10.91%
gcc_ref32_O3	done	13.94%	9.32%	18.06%	10.74%	11.04%	62.89%	12.68%	19.94%
mcf	done	24.46%	11.70%	30.85%	9.98%	5.87%	69.97%	9.12%	20.77%
perlbench_checkspam	done	6.93%	2.21%	10.27%	3.55%	2.91%	19.65%	1.52%	2.59%
perlbench_diffmail	done	8.95%	2.24%	11.37%	12.13%	4.88%	29.10%	2.49%	7.67%
perlbench_splitmail	done	13.58%	1.38%	17.71%	8.34%	7.81%	14.31%	1.51%	2.69%
bwaves_1	done	44.41%	7.12%	48.35%	1.07%	38.47%	90.83%	8.93%	1.22%
bwaves_2	done	44.03%	7.03%	48.06%	1.14%	38.10%	91.21%	8.45%	1.31%
bwaves_3	done	45.97%	7.16%	49.90%	1.06%	41.59%	41.97%	19.35%	1.47%
bwaves_4	done	44.01%	7.08%	47.99%	1.21%	37.57%	44.00%	18.07%	0.98%
exchange2	done	3.11%	1.88%	3.96%	2.01%	2.31%	0.55%	0.15%	0.52%
fotonik3d	done	22.00%	9.79%	28.97%	0.50%	5.16%	49.28%	13.24%	3.95%
roms	done	30.33%	6.83%	37.70%	3.25%	19.59%	40.55%	15.25%	5.82%
wrf	done	26.09%	13.66%	27.65%	8.20%	22.54%	29.03%	13.89%	17.61%
cam4	done	15.12%	5.77%	16.68%	4.62%	15.55%	47.00%	9.17%	18.53%
pooled		15.65%	5.43%	19.22%	4.14%	9.83%	45.16%	14.40%	13.25%
mean		17.31%	6.36%	20.52%	5.34%	13.04%	41.71%	8.68%	10.52%

Miss rates: small-4way (L1D 16 sets x 4, L2 64 sets x 8)

36 of 36 runs completed, 0 running, 0 pending; **0 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.63%	5.18%	33.25%	4.30%	13.06%	64.99%	12.26%	9.77%
parest	done	10.38%	2.36%	10.98%	1.67%	5.06%	70.85%	3.14%	27.35%
deepsjeng	done	8.81%	2.10%	11.44%	7.45%	4.16%	6.35%	2.23%	14.14%
leela	done	4.94%	0.51%	5.49%	0.00%	3.55%	23.48%	0.03%	21.59%
namd	done	11.58%	2.40%	14.75%	1.26%	2.43%	27.50%	7.59%	16.11%
lbm	done	17.63%	0.01%	6.46%	0.12%	75.48%	97.01%	0.06%	12.21%
nab	done	5.40%	0.13%	7.58%	0.69%	0.43%	59.18%	9.30%	1.65%
xz_cld	done	10.66%	2.04%	15.30%	1.51%	2.95%	35.09%	0.03%	0.26%
xz_cpu2006docs	done	9.40%	2.20%	10.33%	1.90%	6.32%	39.60%	4.81%	6.67%
xz_combined	done	9.72%	1.98%	12.37%	1.30%	4.34%	24.47%	0.01%	1.29%
imagick	done	2.18%	3.93%	2.77%	0.03%	0.12%	96.60%	57.37%	19.19%
blender	done	8.00%	2.67%	8.76%	1.42%	3.47%	18.31%	1.21%	46.53%
povray	done	9.16%	7.21%	11.34%	18.52%	3.61%	45.94%	51.80%	63.63%
omnetpp	done	16.31%	0.08%	21.74%	1.50%	5.71%	28.08%	50.69%	2.63%
xalancbmk	done	17.12%	11.80%	19.71%	22.71%	4.54%	70.02%	6.11%	5.53%
x264_pass1	done	2.42%	0.13%	2.87%	1.00%	3.14%	28.58%	6.19%	15.50%
x264_pass2	done	1.95%	0.19%	2.37%	0.78%	2.00%	35.53%	0.85%	18.84%
x264_seek	done	1.96%	0.16%	2.41%	0.91%	1.93%	31.56%	1.11%	13.54%
gcc_pp_O3	done	8.74%	1.30%	11.63%	4.77%	3.64%	41.84%	5.87%	12.58%
gcc_pp_O2	done	8.89%	1.19%	11.84%	4.31%	3.90%	38.46%	6.57%	11.89%
gcc_smaller	done	10.48%	2.05%	13.46%	3.71%	3.44%	47.59%	3.59%	10.17%
gcc_ref32_O5	done	8.86%	1.68%	11.62%	4.07%	3.81%	50.61%	7.41%	12.73%
gcc_ref32_O3	done	10.65%	2.48%	12.31%	5.12%	9.62%	72.66%	4.57%	18.30%
mcf	done	21.53%	6.09%	27.47%	6.44%	4.64%	74.18%	3.56%	19.24%
perlbench_checkspam	done	2.50%	0.10%	3.92%	0.43%	1.13%	28.16%	2.52%	8.95%
perlbench_diffmail	done	4.51%	0.13%	6.17%	4.56%	1.79%	37.92%	0.37%	12.19%
perlbench_splitmail	done	8.88%	0.03%	12.58%	7.27%	2.96%	8.50%	2.60%	0.41%
bwaves_1	done	44.12%	6.22%	48.24%	0.10%	36.83%	23.63%	3.19%	0.58%
bwaves_2	done	43.90%	6.02%	48.01%	0.12%	37.89%	23.88%	3.03%	0.66%
bwaves_3	done	43.19%	6.11%	47.21%	0.00%	36.20%	24.11%	3.53%	67.82%
bwaves_4	done	43.28%	5.84%	47.31%	0.00%	36.77%	23.87%	2.64%	38.55%
exchange2	done	1.67%	1.41%	2.15%	2.24%	0.96%	0.94%	0.09%	0.40%
fotonik3d	done	10.25%	4.03%	14.24%	0.04%	0.18%	66.95%	1.93%	6.94%
roms	done	15.13%	2.31%	18.45%	0.52%	11.85%	53.37%	13.34%	13.66%
wrf	done	11.27%	6.63%	11.52%	4.40%	11.13%	33.77%	8.59%	11.72%
cam4	done	10.69%	2.04%	11.57%	2.35%	12.15%	55.53%	8.09%	25.31%
pooled		10.79%	3.07%	13.38%	2.69%	6.61%	47.07%	10.11%	11.85%
mean		13.11%	2.80%	15.55%	3.26%	10.03%	41.92%	8.23%	15.79%

Miss rates: small-8way (L1D 8 sets x 8, L2 64 sets x 8)

36 of 36 runs completed, 0 running, 0 pending; **0 remaining**.

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.86%	5.46%	33.56%	4.62%	12.90%	64.24%	11.55%	9.11%
parest	done	9.94%	1.37%	10.52%	1.33%	4.88%	73.98%	5.39%	34.21%
deepsjeng	done	8.51%	1.85%	11.04%	7.30%	4.09%	6.50%	2.53%	14.37%
leela	done	4.20%	0.06%	4.76%	0.00%	2.73%	27.29%	0.22%	22.54%
namd	done	11.53%	2.43%	14.69%	1.26%	2.42%	27.58%	7.51%	16.05%
lbm	done	17.45%	0.00%	6.03%	0.10%	75.51%	97.87%	100.00%	14.02%
nab	done	4.13%	0.09%	5.81%	0.26%	0.42%	77.32%	12.94%	4.29%
xz_cld	done	10.24%	1.02%	15.13%	0.48%	2.25%	36.46%	0.05%	0.81%
xz_cpu2006docs	done	9.31%	1.55%	10.31%	1.37%	5.86%	40.00%	6.83%	9.28%
xz_combined	done	9.53%	1.13%	12.39%	0.71%	3.66%	24.91%	0.02%	2.35%
imagick	done	2.19%	3.59%	2.78%	0.01%	0.12%	100.00%	62.71%	45.85%
blender	done	7.97%	2.59%	8.75%	1.31%	3.31%	18.37%	1.24%	50.45%
povray	done	8.45%	7.25%	10.53%	18.27%	3.15%	49.55%	51.51%	64.06%
omnetpp	done	16.27%	0.08%	21.69%	1.19%	5.70%	28.12%	50.69%	3.26%
xalancbmk	done	16.84%	11.17%	19.39%	20.10%	4.47%	71.10%	6.45%	6.24%
x264_pass1	done	2.37%	0.10%	2.80%	0.99%	3.12%	29.18%	8.27%	15.71%
x264_pass2	done	1.83%	0.11%	2.22%	0.76%	1.97%	37.28%	1.30%	19.19%
x264_seek	done	1.87%	0.12%	2.29%	0.86%	1.91%	32.75%	1.36%	14.40%
gcc_pp_O3	done	8.28%	0.85%	11.02%	4.69%	3.48%	43.74%	8.74%	12.71%
gcc_pp_O2	done	8.50%	0.81%	11.31%	4.47%	3.73%	39.97%	8.88%	11.39%
gcc_smaller	done	10.11%	1.46%	12.99%	3.25%	3.32%	49.17%	4.79%	11.57%
gcc_ref32_O5	done	8.29%	1.16%	10.85%	3.63%	3.63%	53.60%	10.84%	14.30%
gcc_ref32_O3	done	10.39%	1.97%	11.75%	5.03%	9.58%	74.28%	4.96%	18.32%
mcf	done	21.44%	5.22%	27.38%	6.19%	4.64%	74.48%	4.06%	20.04%
perlbench_checkspam	done	1.96%	0.09%	3.07%	0.39%	0.87%	34.15%	2.72%	9.90%
perlbench_diffmail	done	4.31%	0.12%	5.90%	6.10%	1.62%	39.09%	0.13%	9.08%
perlbench_splitmail	done	8.75%	0.03%	12.45%	6.70%	2.89%	8.56%	2.96%	0.45%
bwaves_1	done	24.64%	6.55%	25.37%	0.07%	35.29%	34.83%	2.78%	0.83%
bwaves_2	done	24.25%	6.28%	24.90%	0.08%	36.22%	35.39%	2.67%	0.95%
bwaves_3	done	21.04%	10.96%	20.59%	0.00%	37.89%	40.82%	1.81%	94.33%
bwaves_4	done	19.67%	10.48%	19.12%	0.00%	37.67%	43.40%	1.23%	53.65%
exchange2	done	0.96%	1.14%	1.29%	1.80%	0.37%	1.61%	0.12%	0.50%
fotonik3d	done	8.47%	2.73%	11.86%	0.03%	0.02%	81.02%	2.80%	11.65%
roms	done	11.98%	1.78%	14.08%	0.61%	11.40%	67.67%	17.23%	11.63%
wrf	done	6.51%	5.81%	6.48%	4.04%	6.78%	57.32%	9.71%	12.77%
cam4	done	10.43%	1.79%	11.24%	2.13%	12.02%	56.87%	9.17%	27.49%
pooled		9.07%	3.08%	10.99%	2.72%	6.13%	54.37%	9.99%	11.71%
mean		10.24%	2.76%	12.12%	3.06%	9.72%	46.62%	11.84%	18.55%

ref input, full runs

config	completed	running	pending	other	remaining
small-2way	0/15	8	7	0	15
small-fa	0/15	8	7	0	15
small-dm	0/15	8	7	0	15
small-4way	0/15	8	7	0	15
small-8way	0/15	8	7	0	15

Runs not completed in every configuration:

run	small-2way	small-fa	small-dm	small-4way	small-8way
xz_cld	pending	pending	pending	pending	pending
xz_combined	running	running	running	running	running
x264_pass1	running	running	running	running	running
gcc_pp_O3	pending	pending	pending	pending	pending
gcc_pp_O2	pending	pending	pending	pending	pending
gcc_smaller	pending	pending	pending	pending	pending
gcc_ref32_O5	pending	pending	pending	pending	pending
gcc_ref32_O3	pending	pending	pending	pending	pending
mcf	running	running	running	running	running
perlbench_diffmail	running	running	running	running	running
perlbench_splitmail	running	running	running	running	running
bwaves_1	pending	pending	pending	pending	pending
bwaves_2	running	running	running	running	running
bwaves_3	running	running	running	running	running
bwaves_4	running	running	running	running	running

No completed run yet: no percentages.

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.